# PIZZA SALES ANALYSIS

Presented by Vishal Chauhan

# INTRODUCTION

The Pizza Sales Analysis Project leverages SQL for querying and preparing data and Power BI for creating an interactive visualization dashboard. The goal of this project is to derive meaningful insights from pizza sales data, such as revenue trends, top-selling items, category contributions, and customer order behavior. This analysis helps in making informed business decisions and improving overall sales strategy.

# PROJECT OBJECTIVES

To analyze pizza sales data and visualize performance metrics that support data-driven business strategies.

Key Goals:

- Track total revenue, orders, and pizzas sold.
- Identify top-selling pizza types and most preferred categories.
- Analyze revenue trends and order distribution over time.
- Measure average pizzas per order to evaluate customer buying behavior.
- Present insights through a Power BI dashboard for easy interpretation.

# PROBLEMS/ CHALLENGES FACED

### Data Cleaning & Preparation

- The raw dataset contained missing values and inconsistent formats (e.g., different date/time formats).
- Some fields had duplicate records, which affected accuracy.
- Columns such as price and quantity needed to be converted to numeric types for calculations.
- Had to ensure proper relationships between multiple tables (orders, order_details, pizzas, pizza_types).

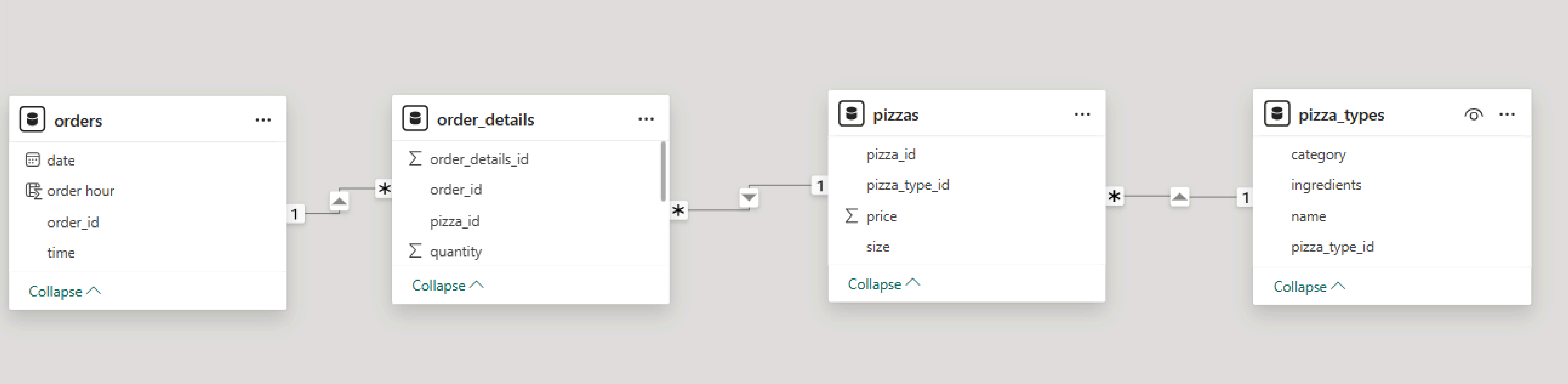
### SQL Query Challenges

- Writing complex joins across multiple tables to fetch meaningful insights.
- Implementing aggregate functions (SUM, COUNT, AVG) with GROUP BY correctly.
- Handling date/time functions that behaved differently in SQL Server vs PostgreSQL.
- Using subqueries and window functions for ranking and cumulative totals.

### Power BI Integration Issues

- Difficulty in importing large CSV files and ensuring data types matched correctly.
- Relationships between tables had to be carefully defined (e.g., one-to-many joins).
- Creating DAX measures (like Running Total, Average Pizzas per Order) required trial and testing.
- Formatting visuals and aligning theme colors for a professional look.

# MODEL VIEW / DATA MODEL




✅ IMPLEMENTED A STAR SCHEMA WITH ONE FACT TABLE (ORDER_DETAILS) AND THREE DIMENSION TABLES (ORDERS, PIZZAS, PIZZA_TYPES).

✅ ENSURED ONE-TO-MANY RELATIONSHIPS FOR ACCURATE DATA AGGREGATION.

✅ USED THIS MODEL TO CREATE DAX MEASURES AND VISUALIZE PERFORMANCE IN POWER BI DASHBOARD.

# RETRIEVE THE TOTAL NUMBER OF ORDER PLACED

```sql
SELECT COUNT(order_id) AS total_orders FROM orders
```

Results | Messages

| | total_orders |
|---|---|
| 1 | 21350 |

# CALCULATE TOTAL REVENUE GENERATED FROM PIZZA SALES

```sql
SELECT
ROUND(SUM(quantity * price),2) AS total_revenue
FROM order_details od
JOIN pizzas ON pizzas.pizza_id = od.pizza_id
```

Results Messages

| | total_revenue |
|---|---|
| 1 | 817860.05 |

# INDENTIFY THE MOST COMMON PIZZA SIZE ORDERED

```sql
SELECT size, COUNT(order_details_id) AS order_count
FROM pizzas p
JOIN order_details od on p.pizza_id = od.pizza_id
GROUP BY size order by order_count desc
```

Results Messages

| | size | order_count |
|---|---|---|
| 1 | L | 18526 |
| 2 | M | 15385 |
| 3 | S | 14137 |
| 4 | XL | 544 |
| 5 | XXL | 28 |

# ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME

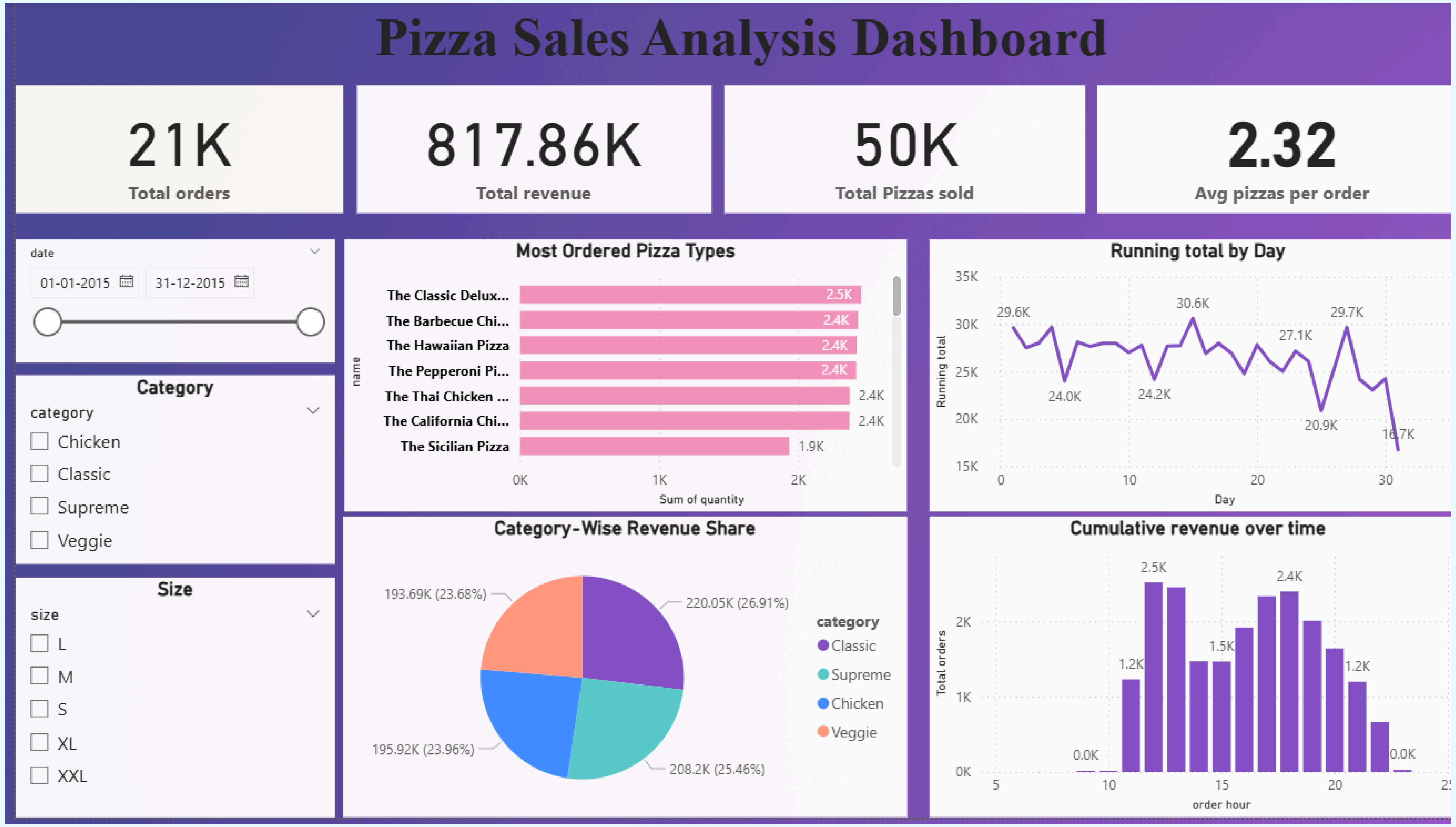
```
SELECT date, round(sum(revenue) OVER(order by date), 2) AS cum_revenu
FROM
(SELECT o.date, SUM(od.quantity * p.price) As revenue
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id
JOIN orders o ON o.order_id = od.order_id
GROUP BY o.date) as sales
```

Results | Messages

| | date | cum_reven |
|---|---|---|
| 1 | 2015-01-01 | 2713.85 |
| 2 | 2015-01-02 | 5445.75 |
| 3 | 2015-01-03 | 8108.15 |
| 4 | 2015-01-04 | 9863.6 |
| 5 | 2015-01-05 | 11929.55 |
| 6 | 2015-01-06 | 14358.5 |
| 7 | 2015-01-07 | 16560.7 |
| 8 | 2015-01-08 | 19399.05 |
| 9 | 2015-01-09 | 21526.4 |
| 10 | 2015-01-10 | 23990.35 |
| 11 | 2015-01-11 | 25862.65 |

# DASHBOARD OVERVIEW

**"THIS IS THE MAIN DASHBOARD, WHERE I TRACKED ALL THE KEY METRICS AT A GLANCE — TOTAL REVENUE, TOTAL ORDERS, TOTAL PIZZAS SOLD, AND THE AVERAGE PIZZAS PER ORDER."**



# CONCLUSION

- The Pizza Sales Analysis dashboard provided a comprehensive understanding of sales performance and customer preferences. By leveraging SQL for data cleaning and transformation and Power BI for visualization, meaningful insights were uncovered.

- The analysis revealed that Classic pizzas contributed the highest share of total revenue, while evening hours saw the greatest number of orders. Additionally, a few top-selling pizza types—such as Barbecue Chicken and Classic Deluxe—were identified as key revenue drivers.

- Overall, this project successfully demonstrated how data-driven decision-making can help businesses identify best-performing products, optimize inventory, and plan targeted marketing strategies.

- Future enhancements can include analyzing customer demographics, regional trends, and seasonal sales patterns to gain deeper business insights

# THANK YOU